



UNIVERSITY OF
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S2 Advanced Statistical Methods for Data Science - Coursework Assignment

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Introduction

The Antikythera Mechanism is one of the most remarkable technological discoveries from the ancient world. Found in 1901 (by accident) in a shipwreck near the Greek island of Antikythera, the device dates back to between 140 and 60 BCE (perhaps another opportunity for bayesian inference). What makes it so extraordinary is its system of bronze gears, which could be manually turned to display a wide range of astronomical phenomena. Housed in a wooden box, the mechanism likely showed the positions of the Sun and Moon, lunar phases, eclipse predictions, and even events tied to the ancient Greek calendar such as the Olympic Games [1].

Thanks techniques like X-ray tomography, researchers have been able to reconstruct parts of the mechanism and get a clearer picture of how it worked. One key component is the calendar ring — a circular disk punctured with holes arranged around its edge. Only part of this ring survives, but it provides clues to the type of calendar the device may have used. While it was once thought to represent a 365-day solar calendar, new analysis suggests it may instead follow a 354-day lunar cycle [2].

In this project, I use Bayesian Inference to infer how many holes the full ring originally contained? Using the known positions of 81 holes across eight broken and misaligned fragments, I apply Hamiltonian Monte Carlo Markov Chains to estimate the total number of holes. Two different models are used to account for uncertainties in the hole positions: one assuming isotropic Gaussian error, and another that separates radial and tangential uncertainties [3]. By comparing the results from both models, I aim to determine which best fits the data and what this can tell us about how the mechanism was designed and about the way time was measured in the ancient world.

1 Measured Hole Locations $d_i \in \mathbb{R}^2$ in the x - y Plane

The data used in this analysis was obtained from the public dataset provided by Thoeni, Budiselic, and Ramsey [4]. The dataset contains the x and y coordinates of 81 holes in

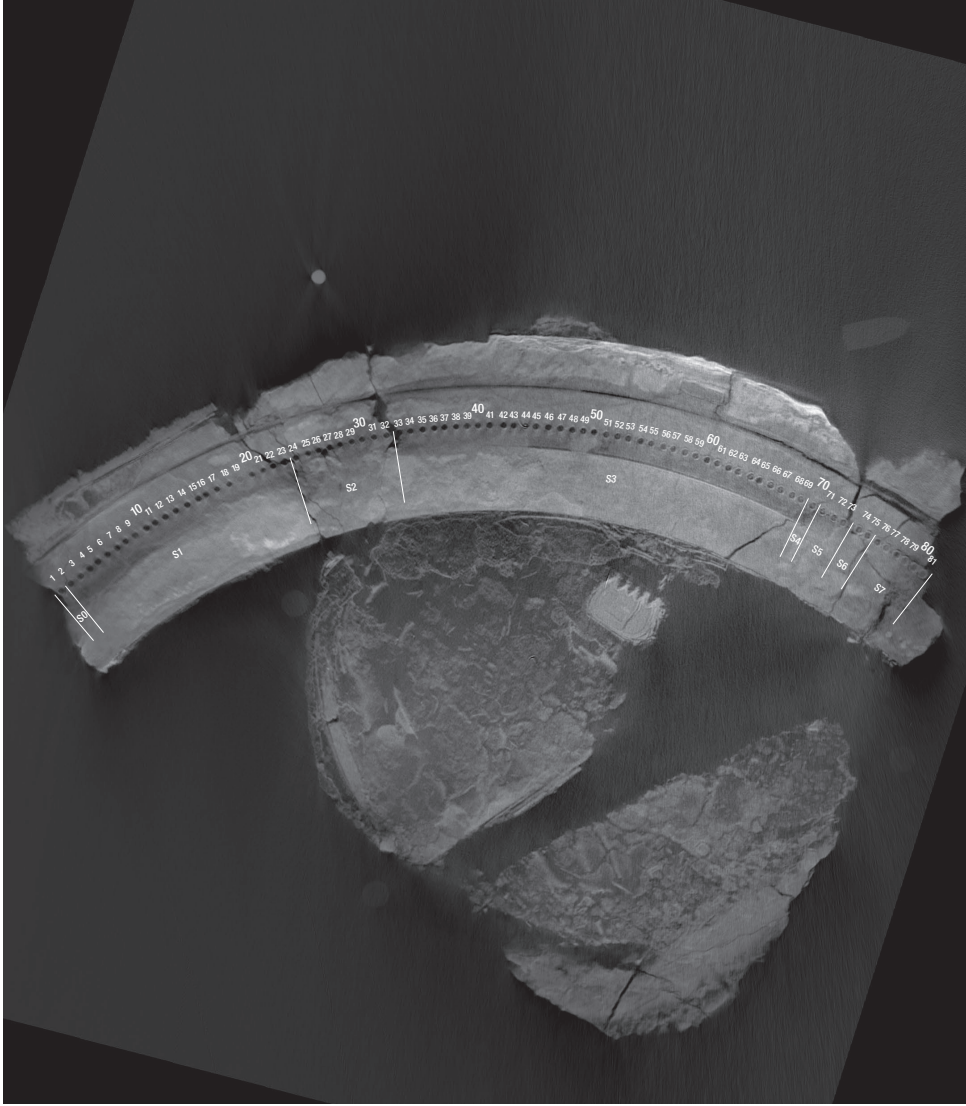


Figure 1: X-ray image of the surviving fragments of the Antikythera Mechanism.

the surviving fragments of the calendar ring. These holes are distributed across eight fragments, each of which is colour-coded in Figure 2.

2 Hole Location model

Following the model described in Section 2 of Woan and Bayley [3], we assume that the original calendar ring contained N holes, evenly spaced around a circle of radius r . The full ring no longer survives, and the measured hole locations come from 8 fragmented sections. Each fragment is slightly misaligned due to relative translations and in-plane rotations.

The Cartesian coordinates of the i -th hole in a given section can be modelled as:

$$\vec{m}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix} = \begin{pmatrix} r \cos \left(\frac{2\pi(i-1)}{N} + \alpha_j \right) + x_{0j} \\ r \sin \left(\frac{2\pi(i-1)}{N} + \alpha_j \right) + y_{0j} \end{pmatrix}, \quad (1)$$

where (x_{0j}, y_{0j}) is the translation of section j , and α_j is its relative angular offset. The

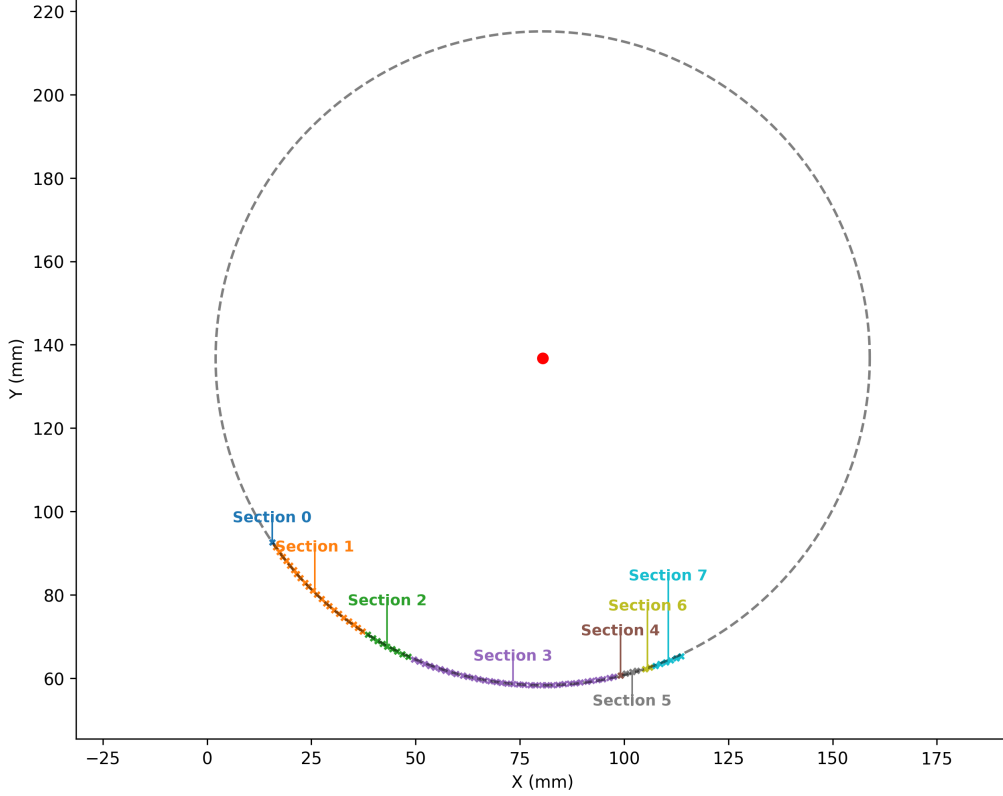


Figure 2: Measured hole locations $d_i \in \mathbb{R}^2$ in the x - y plane, colour-coded by fractured ring segment. The dashed circle shows the estimated full ring, fitted to the data using a non-linear least-squares method. Section labels are placed to minimise overlap and connected to the median point of each segment.

model assumes all fragments lie in the same x - y plane and are internally undistorted.

3 Likelihood Model

To assess how well a proposed model predicts the observed hole positions, we assume that each measured hole location $\vec{d}_i = (x_i, y_i)$ is an independent realisation from a two-dimensional Gaussian distribution centred on a predicted location $\vec{m}_i = (x_i^{\text{model}}, y_i^{\text{model}})$. The set of all measurements is denoted $D = \{\vec{d}_i \mid i = 1, \dots, N_{\text{holes}}\}$, with $N_{\text{holes}} = 81$.

3.1 General Form of the Likelihood

Assuming independent Gaussian errors, the likelihood for a single hole is:

$$p(\vec{d}_i \mid \vec{m}_i, \Sigma) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\vec{d}_i - \vec{m}_i)^\top \Sigma^{-1}(\vec{d}_i - \vec{m}_i)\right), \quad (2)$$

where Σ is the 2×2 covariance matrix characterising the uncertainty. Aggregating over all holes, the full log-likelihood is:

$$\log \mathcal{L}(D \mid \theta) = -\frac{1}{2} \sum_{i=1}^{N_{\text{holes}}} \left[(\vec{d}_i - \vec{m}_i)^\top \Sigma^{-1}(\vec{d}_i - \vec{m}_i) + \log(2\pi|\Sigma|) \right], \quad (3)$$

where θ denotes the full set of model parameters:

$$\theta = \{N, r, \text{error parameters in } \Sigma, (x_j, y_j, \alpha_j) \text{ for } j = 0, \dots, 7\}.$$

3.2 Isotropic Gaussian Error Model

In the isotropic case, errors in x and y directions are assumed equal and uncorrelated:

$$\Sigma = \sigma^2 I, \quad \Sigma^{-1} = \frac{1}{\sigma^2} I, \quad |\Sigma| = \sigma^4, \quad (4)$$

with a single noise parameter σ . Substituting into the log-likelihood yields:

$$\log \mathcal{L}(D \mid \theta) = -\frac{1}{2\sigma^2} \sum_{i=1}^{N_{\text{holes}}} \left\| \vec{d}_i - \vec{m}_i \right\|^2 - \frac{N_{\text{holes}}}{2} \log(2\pi\sigma^2). \quad (5)$$

3.3 Anisotropic Radial–Tangential Error Model

To account for direction-dependent uncertainty, we define a local radial–tangential frame for each hole. Given angular position ϕ_i , the unit vectors are:

$$\hat{r}_i = \begin{pmatrix} \cos \phi_i \\ \sin \phi_i \end{pmatrix}, \quad \hat{t}_i = \begin{pmatrix} -\sin \phi_i \\ \cos \phi_i \end{pmatrix}. \quad (6)$$

The displacement between predicted and observed positions is:

$$\vec{\delta}_i = \vec{d}_i - \vec{m}_i. \quad (7)$$

Projected onto the local basis:

$$\delta_{r,i} = \hat{r}_i^\top \vec{\delta}_i, \quad \delta_{t,i} = \hat{t}_i^\top \vec{\delta}_i. \quad (8)$$

Mahalanobis Distance and Log-Likelihood

The Mahalanobis distance generalises Euclidean distance by incorporating directional uncertainty:

$$D_i^2 = \begin{pmatrix} \delta_{r,i} \\ \delta_{t,i} \end{pmatrix}^\top \begin{pmatrix} 1/\sigma_r^2 & 0 \\ 0 & 1/\sigma_t^2 \end{pmatrix} \begin{pmatrix} \delta_{r,i} \\ \delta_{t,i} \end{pmatrix} = \frac{\delta_{r,i}^2}{\sigma_r^2} + \frac{\delta_{t,i}^2}{\sigma_t^2}. \quad (9)$$

This distance penalises deviations more strongly in directions of higher precision (smaller σ). It replaces the quadratic form in Equation (2) for the anisotropic case, leading to:

$$\log \mathcal{L}(D \mid \theta) = -\frac{1}{2} \sum_{i=1}^{N_{\text{holes}}} [D_i^2 + \log(2\pi\sigma_r\sigma_t)], \quad (10)$$

where the covariance determinant is $|\Sigma_{\text{rt}}| = \sigma_r^2 \sigma_t^2$.

3.4 Parameter Summary

All parameters influence the likelihood in one of three ways:

- N and r define the angular placement and radius of holes around the full ring.
- (x_j, y_j, α_j) describe the translation and rotation of each fractured segment.
- σ (isotropic) or (σ_r, σ_t) (anisotropic) define the scale of measurement uncertainty and influence both the distance and normalisation terms in the log-likelihood.

Both error models were implemented in NumPy and JAX. The JAX version supports automatic differentiation and was used for posterior sampling with Hamiltonian Monte Carlo via NumPyro. Consistency between implementations was verified numerically.

4 Derivatives of Log-Likelihood

We computed its gradient with respect to the model parameters θ_μ :

$$\frac{\partial}{\partial \theta_\mu} \log L(\mathcal{D} \mid \boldsymbol{\theta})$$

Rather than deriving analytic expressions for the derivatives, we opted to use automatic differentiation via the jax library. The log-likelihood was implemented in a fully JAX-compatible format, which allowed for straightforward computation of the gradient and Hessian. Automatic differentiation works by applying the chain rule systematically (as shown in the previous Research Computing C1 coursework) to the sequence of operations used to evaluate the function, allowing exact derivatives to be computed efficiently and accurately without symbolic algebra or numerical approximation.

To ensure the correctness of the computed gradients, we also implemented a finite difference approximation using central differences:

$$\frac{\partial \log L}{\partial \theta_\mu} \approx \frac{\log L(\theta_\mu + \epsilon) - \log L(\theta_\mu - \epsilon)}{2\epsilon}$$

This was implemented using NumPy and compared against the JAX gradients for all parameters. To test the gradient computation independently of model fitting, we evaluated both methods at a set of arbitrary parameter values that we selected manually. This allowed us to verify that the automatic differentiation mechanism worked correctly across a region of parameter space.

We then computed the relative error between the two methods:

$$\text{Relative Difference} = \frac{|\nabla_{\text{JAX}} - \nabla_{\text{FD}}|}{|\nabla_{\text{FD}}| + \epsilon}$$

The results were visualised in a bar chart with a logarithmic y-axis, shown in Figure 3. The relative differences were found to be negligible (on the order of 10^{-6} or less), confirming the accuracy of the automatic differentiation approach.

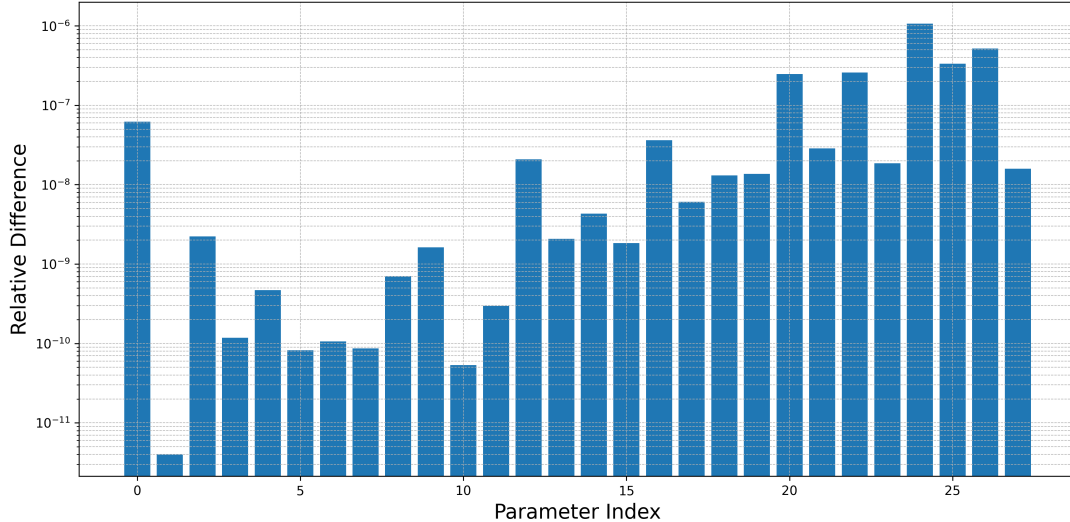


Figure 3: Relative difference between gradients computed via JAX autodiff and finite differences. Each bar corresponds to a parameter θ_μ .

Given the simplicity and clarity of using `jax`, as well as the reliability of the results validated via finite differences, we chose to use automatic differentiation for all tasks requiring the computation of gradients.

5 Maximum Likelihood Estimation

To estimate the best-fitting model parameters, we performed maximum likelihood estimation (MLE) for both the isotropic and anisotropic covariance models. Sections 0 and 4, which each contained only a single hole, were excluded. The remaining data—79 holes across six sections—were sorted by section and hole number, and grouped into (x, y) coordinate lists for each section.

We used the `L-BFGS-B` optimisation algorithm [5] to maximise the log-likelihood function, with gradients computed via automatic differentiation using `jax`. Optimisation was carried out on the negative log-likelihood.

For the anisotropic model, the parameters include:

- Total number of holes N
- Ring radius R
- Radial and tangential uncertainties σ_r and σ_t
- Angular offsets ϕ_j , and x/y centre offsets for each of the six sections

The parameter ordering is shown in Table 1.

Index	Parameter
0	N (number of holes)
1	R (ring radius)
2	σ_r (radial uncertainty)
3	σ_t (tangential uncertainty)
4–9	ϕ_1 to ϕ_6 (angular offsets)
10–15	x_1 to x_6 (section x offsets)
16–21	y_1 to y_6 (section y offsets)

Table 1: Parameter ordering for the anisotropic model.

Initial values were based on dataset medians with small perturbations added to avoid local minima. Parameter bounds were applied to ensure physical validity (e.g., $N > 0$, $\sigma > 0$).

The optimiser returned the following result:

Maximised Log-Likelihood: 230.67

```
[ 3.55138326e+02  7.73198713e+01  2.57642786e-02  1.22578786e-01
-2.52567532e+00 -2.13565124e+00 -1.97423759e+00 -1.32147032e+00
-1.26317302e+00 -1.25305939e+00  7.96772149e+01  7.98973406e+01
 7.98583275e+01  8.13733544e+01  8.15575431e+01  8.32199045e+01
 1.36019135e+02  1.35701360e+02  1.35689755e+02  1.36066927e+02
 1.35837969e+02  1.36401869e+02]
```

Using these parameters, we predicted the expected hole positions for each section and overlaid them on the measured hole locations (Figure 4). To reduce clutter, the predicted points were vertically shifted by 10 mm and arrows were added from the shifted locations to their true positions. The same process was repeated for the isotropic model.

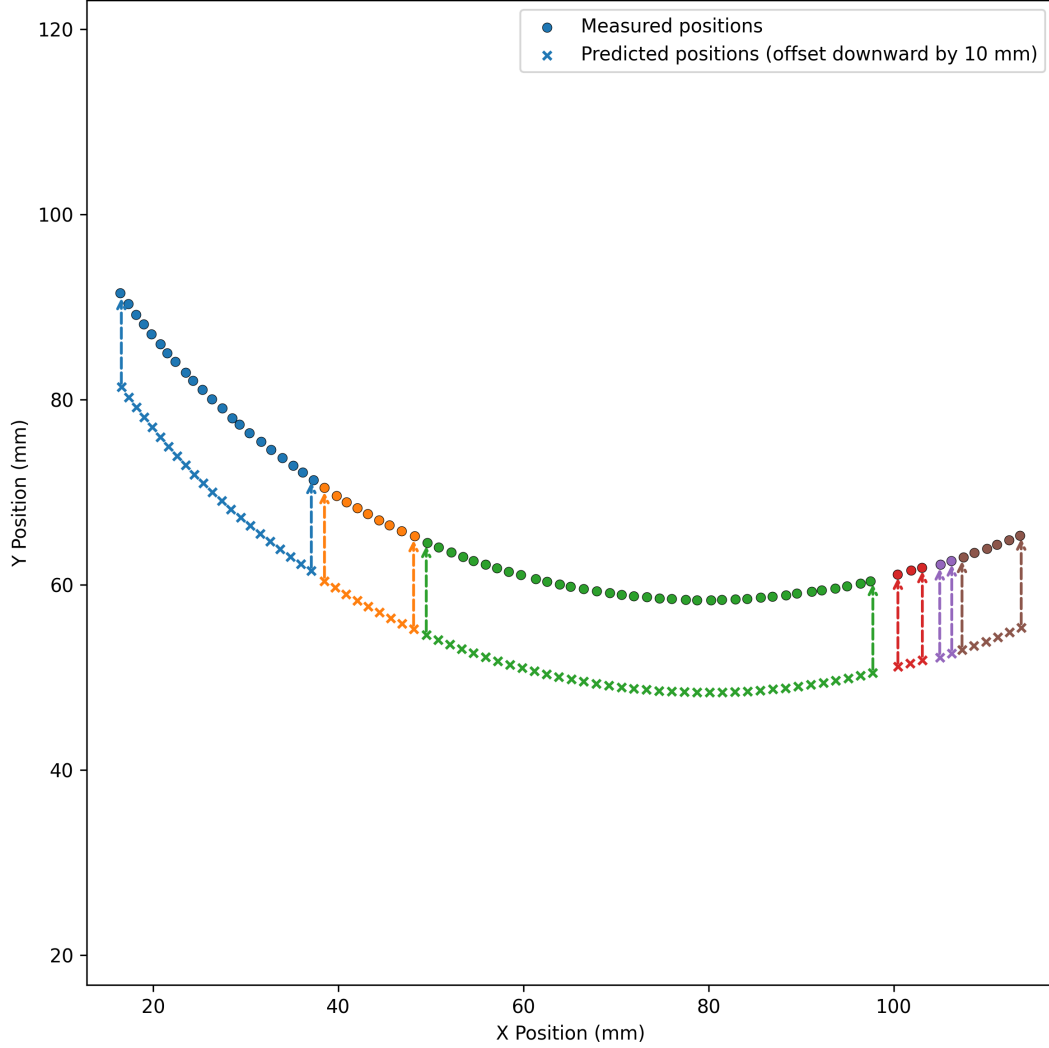


Figure 4: Measured hole positions (circles) overlaid with predicted positions from the anisotropic model (x markers), vertically shifted by 10 mm for clarity. Arrows connect the first and last predicted points in each section to their true model locations. MLE estimation: $N = 355.14$, $R = 77.32$.

We repeated the optimisation process for the isotropic model, where a single shared uncertainty parameter σ is used for both radial and tangential directions. The same objective function, data format, and parameter initialisation strategy were used, with the exception that only one σ term was included.

The optimiser returned the following result:

Maximised Log-Likelihood: -20.37

```
[ 3.54953104e+02  7.72779993e+01  1.24925618e-01 -2.52545406e+00
 -2.13497172e+00 -1.97450722e+00 -1.31829739e+00 -1.26683207e+00
 -1.25314722e+00  7.96396936e+01  7.98280318e+01  7.98622123e+01
 8.11471212e+01  8.18393995e+01  8.32405652e+01  1.36016332e+02
 1.35689853e+02  1.35638898e+02  1.35961123e+02  1.35884849e+02
 1.36361957e+02]
```

As before, we predicted the hole positions using the optimised parameters and overlaid them on the measured data (Figure 5). The same plotting convention was used:

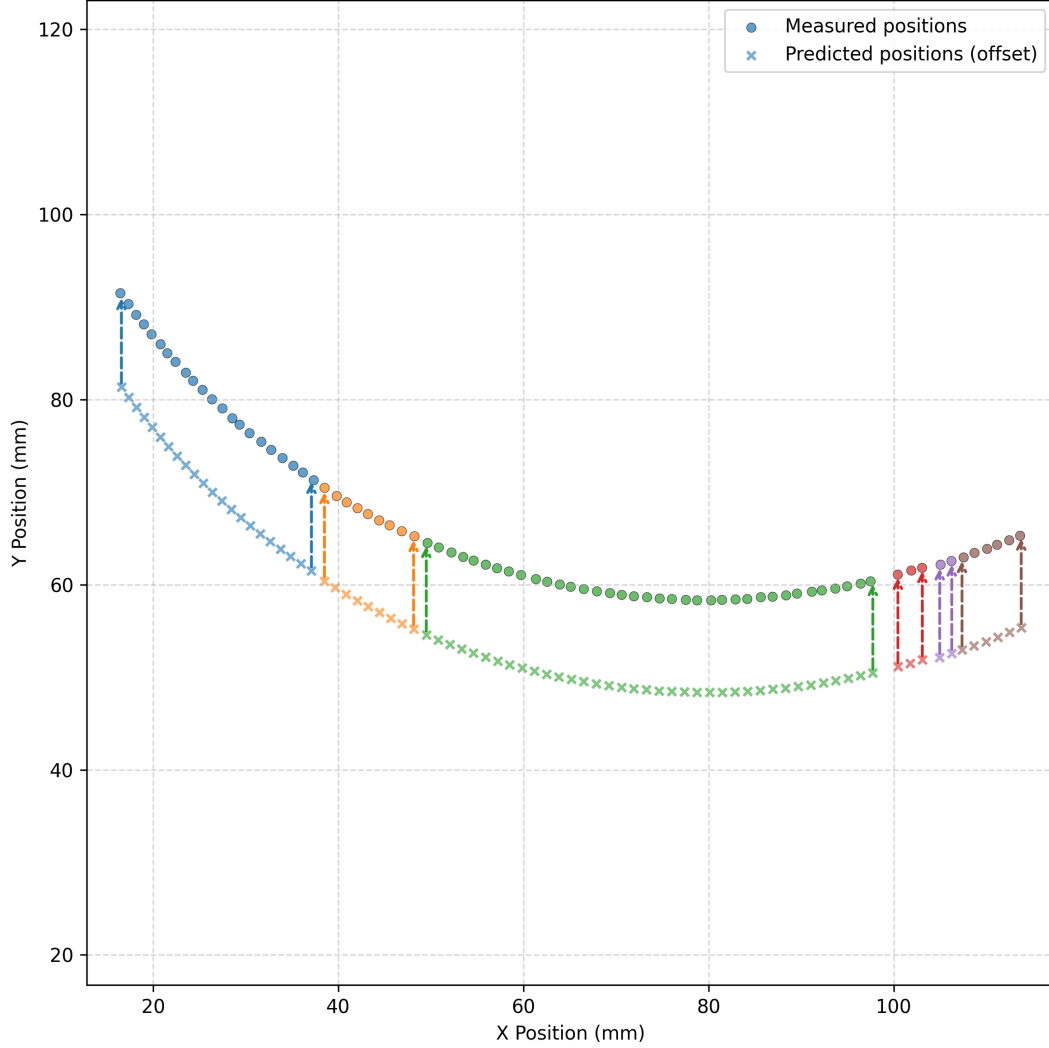


Figure 5: Measured hole positions (circles) overlaid with predicted positions from the isotropic model (x markers), vertically shifted by 10 mm for clarity. Arrows connect the first and last predicted points in each section to their true model locations. MLE estimation: $N = 354.95$, $R = 77.38$.

Comparing the two models, the anisotropic version achieved a higher maximised log-likelihood ($\log \mathcal{L} = 230.67$) compared to the isotropic model ($\log \mathcal{L} = -20.37$). This difference suggests that allowing separate treatment of radial and tangential uncertainties provides a better fit to the measured hole positions.

6 Bayesian Inference and Posterior Sampling

To refine our understanding of the hole positions, we performed Bayesian inference using Hamiltonian Monte Carlo (HMC) sampling, implemented via the `NumPyro` library. This approach allowed us to estimate the full posterior distributions of the model parameters, incorporating both prior knowledge and the observed data.

The priors were informed by maximum likelihood estimates (MLE), previous literature, and physical considerations—particularly the hypothesis that the mechanism tracks a lunar calendar. These priors are summarised below and visualised in Figure 6:

- **Total number of holes:** $N \sim \mathcal{N}(354.4, 3.5)$, reflecting the hypothesis that the calendar ring encodes a lunar year of 12 months (each ≈ 29.53 days). The mean is informed by our own maximum likelihood estimate (MLE), as well as historical interpretations of the mechanism’s purpose as a calendrical device [6][7][8].
- **Ring radius:** $R \sim \mathcal{N}(77.35, 2)$, based on the MLE results and results published by Woan and Bayley [3].
- **Uncertainties:**
 - For the *anisotropic model*, radial and tangential uncertainties were given log-normal priors to reflect the fact that uncertainties cannot be negative. The peaks of these priors also align with results from our MLE and results published by Woan and Bayley [3].

$$\sigma_r \sim \text{LogNormal}(\log(0.028), 0.25), \quad \sigma_t \sim \text{LogNormal}(\log(0.13), 0.3).$$

- For the *isotropic model*, the shared uncertainty parameter was:

$$\sigma \sim \text{LogNormal}(\log(0.13), 0.45).$$

- **Section-specific parameters:** angular offsets ϕ_j , and spatial offsets x_j, y_j . Again, these were chosen to align with our MLE results as well as results published by Woan and Bayley [3]. These were given a large standard deviation to allow our HMC to explore a broad parameter space looking at within-section misalignments and minor translational shifts between fractured segments:

$$\phi_j \sim \mathcal{N}(-2.5, 0.5), \quad x_j \sim \mathcal{N}(80, 5), \quad y_j \sim \mathcal{N}(135, 5).$$

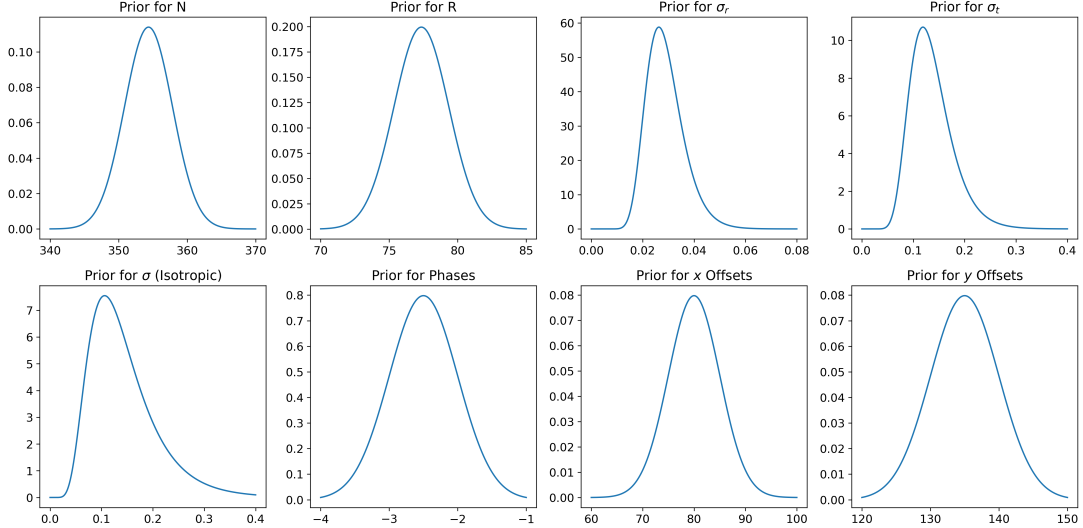


Figure 6: Visualisation of the prior distributions for the model parameters. Each subplot corresponds to one of the priors described above.

Posterior sampling was performed using the No-U-Turn Sampler (NUTS) [9], a variant of HMC known for its efficiency in high-dimensional spaces. We used 12 parallel chains (corresponding to the number of CPU cores on my machine), each with 10,000 warmup steps and 50,000 sampling steps. Initialisation was done via the `init_to_median` strategy for robust convergence.

Anisotropic Model Results

For the anisotropic model, trace plots for key parameters (Figure 7) show good mixing and convergence across all chains, with minimal autocorrelation. The posterior distributions and joint correlations between parameters are shown in the corner plot in Figure 8.

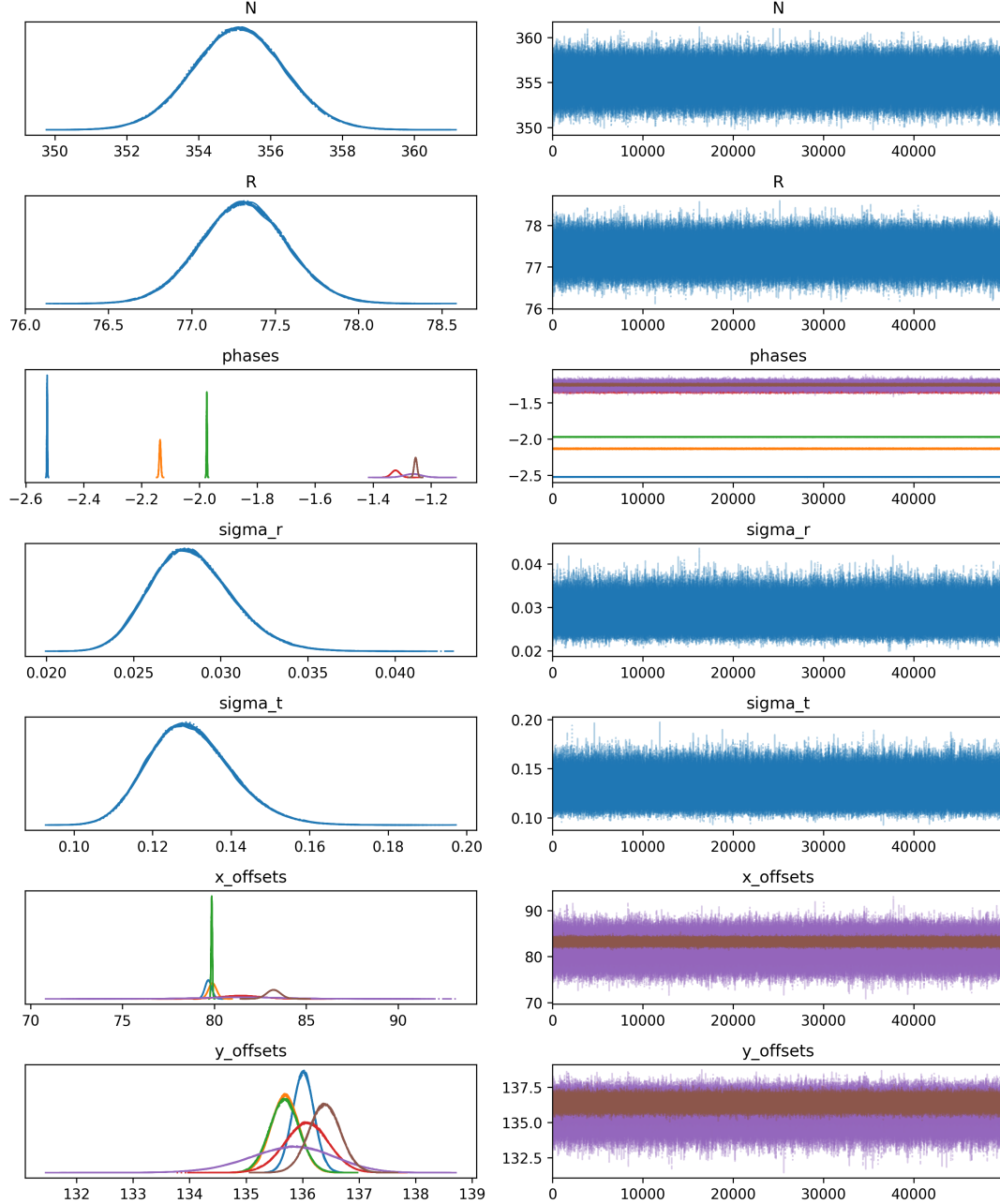


Figure 7: Trace plots for the anisotropic model. Each colour represents one of the 12 chains.

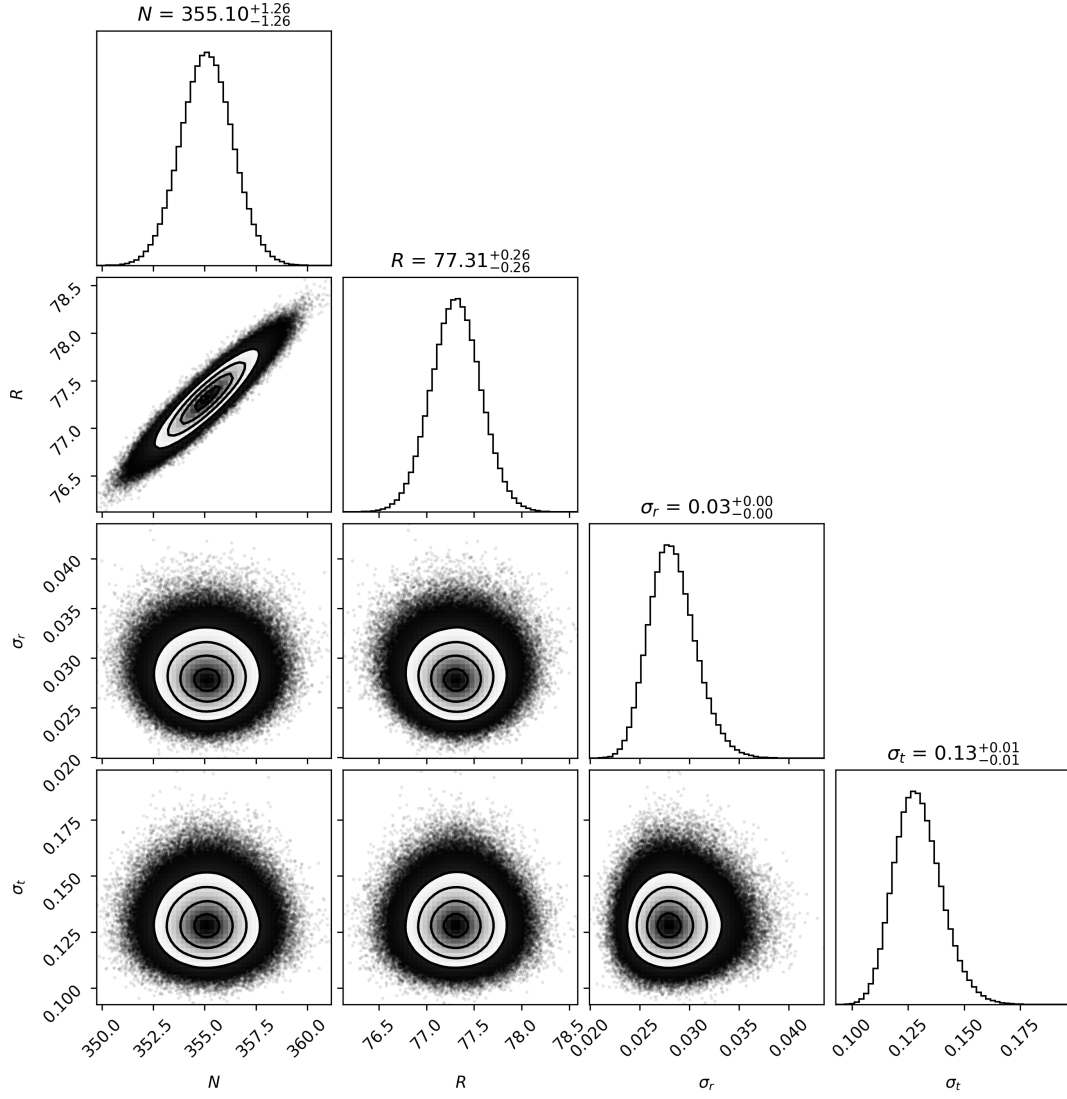


Figure 8: Corner plot showing posterior distributions and joint correlations for the anisotropic model parameters N , R , σ_r , and σ_t .

Isotropic Model Results

For comparison, the isotropic model was fit using the same MCMC settings. The trace plots (Figure 9) again demonstrate good convergence, and the corresponding corner plot (Figure 10) summarises the posterior distributions and joint parameter behaviour.

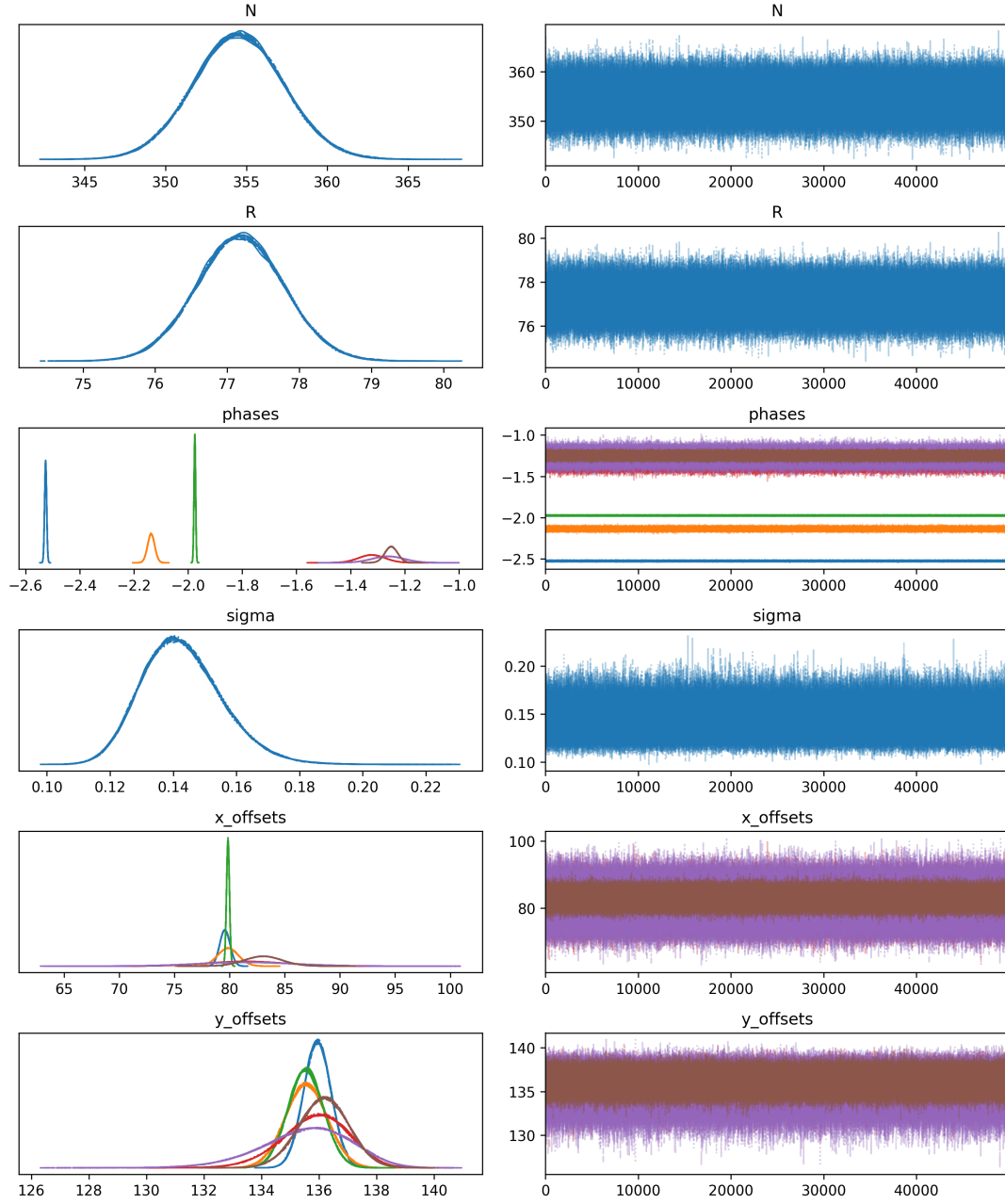


Figure 9: Trace plots for the isotropic model parameters across 12 chains.

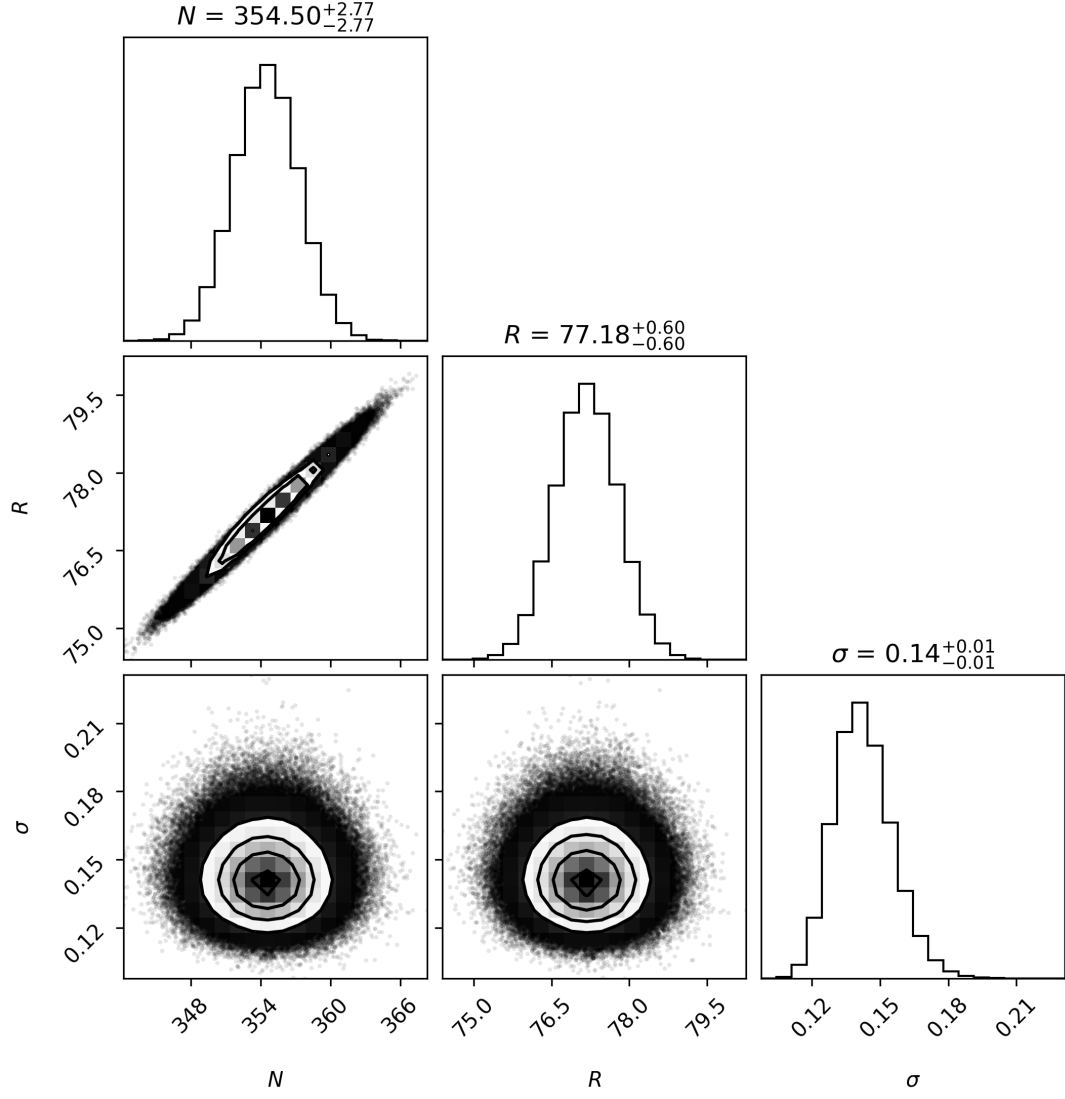
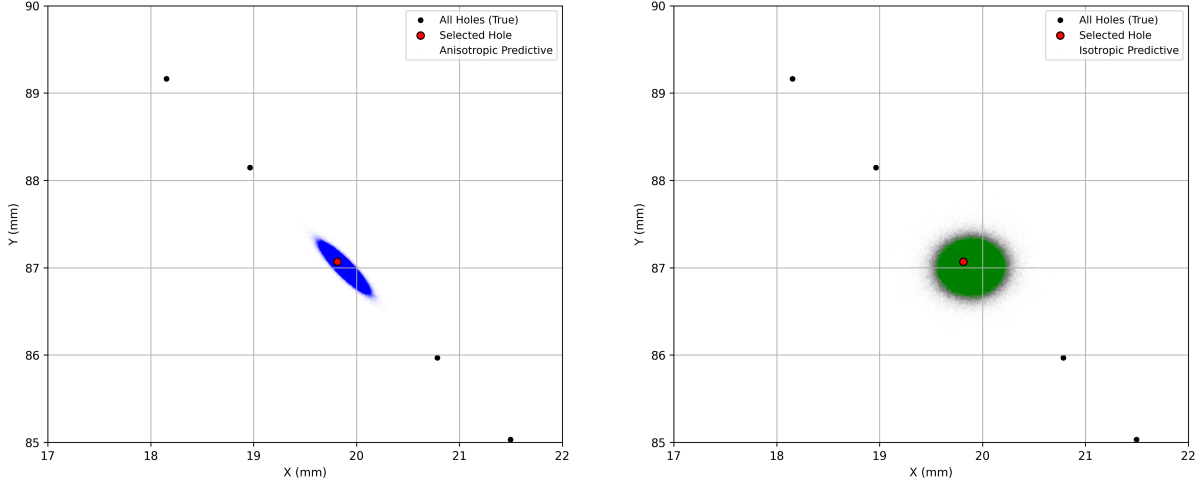


Figure 10: Corner plot showing the posterior distributions and pairwise correlations for the isotropic model parameters: N , R , and σ .



(a) Anisotropic model

(b) Isotropic model

Figure 11: Posterior predictive distributions for the position of a single hole (Section 1, Hole 5) under the anisotropic and isotropic models. Black dots show all holes in the selected section; the red dot marks the true location of the selected hole. Blue and green dots represent posterior predictive samples under each model.

The predictive distributions show clear differences between the two models. In the isotropic case (Figure 11b), the posterior predictive samples form a near-circular cloud around the true location, which is consistent with the assumption of equal measurement uncertainty in all directions.

The anisotropic model (Figure 11a) produces a clearly elliptical distribution. This shape reflects the underlying assumption that uncertainty differs between the radial and tangential directions, given the circular structure of the mechanism. Radial positions are better constrained due to the curvature of the ring, while tangential offsets are more sensitive to misalignments between broken fragments. This results in a tight ellipse along the radial direction and a broader spread tangentially.

The sharpness and clarity of these distributions are a direct result of the large number of posterior samples used. We ran 12 parallel chains, each drawing 50,000 samples after 10,000 warmup steps, producing 600,000 raw samples.

7 Covariance Structure and Model Comparison

The covariance matrix Σ plays a central role in modelling uncertainty in the measured hole positions. In two dimensions, Σ defines the shape and orientation of the uncertainty ellipse around each predicted hole location $\vec{m}_i$, encoding not only the variance in the x and y directions but also potential correlation between them. In the isotropic case, the same uncertainty is assumed in all directions, leading to circular error regions:

$$\Sigma = \sigma^2 I.$$

However, this may be overly simplistic. A more physically realistic model accounts for different uncertainty along the radial and tangential directions of the ring:

$$\Sigma_{\text{rt}} = \begin{pmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_t^2 \end{pmatrix}$$

in the local radial–tangential coordinate frame. This anisotropic model reflects the idea that measurement uncertainty is typically larger in the direction *around* the ring (tangential) than *towards/away from* the centre (radial), especially when holes are arc-fitted or distorted by curvature.

Model Comparison

To formally assess whether the data favour the anisotropic model, we used the Savage-Dickey density ratio to compute the Bayes factor between the isotropic and anisotropic models. This method is valid when one model (here, the isotropic model) is a special case of the other (the anisotropic model), obtained by setting a specific parameter to a fixed value.

In our case, the isotropic model corresponds to $\Delta = \sigma_t - \sigma_r = 0$, where the radial and tangential uncertainties are equal. The Savage-Dickey ratio compares the posterior and prior densities of this parameter at $\Delta = 0$:

$$\text{Bayes Factor} = \frac{p(\Delta = 0 \mid D)}{p(\Delta = 0)}.$$

We estimated these densities using kernel density estimation (KDE) applied to posterior and prior samples of Δ . The posterior placed almost no probability mass at $\Delta = 0$, whereas the prior allowed moderate support:

Posterior density at $\Delta = 0$: 0.00000, Prior: 0.00868,

$\Rightarrow$ Bayes Factor (isotropic vs. anisotropic) ≈ 0.000 .

This result provides decisive evidence *against* the isotropic model. The data strongly favour a model where radial and tangential uncertainties differ — a physically reasonable outcome, given the geometry of the circular ring. The anisotropic model better captures the direction-dependent structure of measurement error.

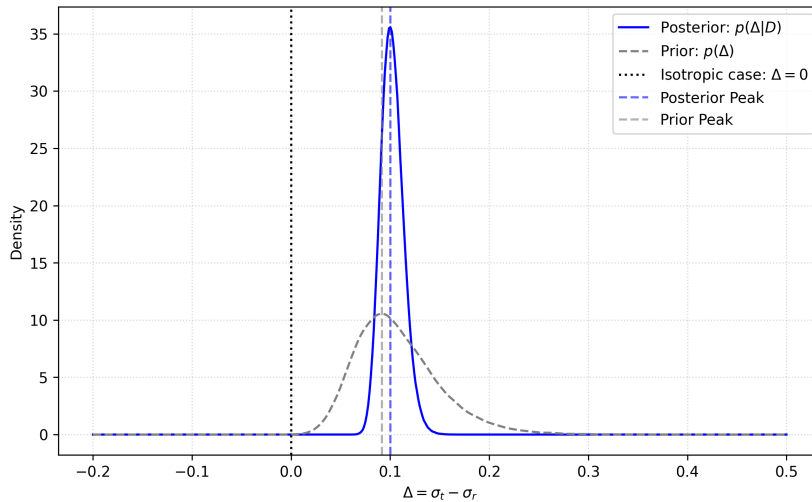


Figure 12: Savage-Dickey density ratio comparing prior and posterior densities of $\Delta = \sigma_t - \sigma_r$. The posterior distribution is sharply peaked away from zero, indicating strong support for the anisotropic model.

This result provides decisive evidence *against* the isotropic model. The data favour a model in which radial and tangential uncertainties differ — a physically reasonable outcome given the mechanism’s circular geometry. The radial–tangential error structure likely arises from the way the ring was measured or manufactured, with uncertainty elongated along the arc.

Nested Sampling and Log-Evidence Comparison

To further assess model preference, we used *nested sampling* to estimate the marginal likelihood (also called the *evidence*) for each model. Unlike the Savage-Dickey ratio, which only compares posterior and prior densities at a single point, nested sampling provides a global measure of how well a model explains the data across its entire parameter space.

The evidence for a model $\mathcal{M}$ is defined as:

$$\log Z = \log \int \mathcal{L}(\theta) \pi(\theta) d\theta,$$

where $\mathcal{L}(\theta)$ is the likelihood and $\pi(\theta)$ is the prior. A higher evidence indicates a better balance between goodness-of-fit and model complexity.

Nested sampling proceeds by drawing a set of live points from the prior and iteratively replacing the worst-fitting point with a better one, gradually focusing on regions of high likelihood. This allows both the evidence and the posterior to be estimated simultaneously.

In our implementation, we used the `dynesty` library with 500 live points and the ‘rwalk’ sampling method. The log-likelihood function was compiled using `jax.jit` for speed, and the prior transform was matched to the priors used in our Bayesian inference (e.g., normal priors for N and R , log-normal priors for σ_r and σ_t).

The resulting log-evidence estimates were:

$$\log Z_{\text{aniso}} = 126.13 \pm 0.70, \quad \log Z_{\text{iso}} = -141.80 \pm 0.71.$$

The difference in log-evidence,

$$\Delta \log Z = \log Z_{\text{aniso}} - \log Z_{\text{iso}} \approx 267.93,$$

is overwhelmingly in favour of the anisotropic model. According to standard interpretive thresholds, a log-evidence difference greater than 10 is considered “very strong” evidence - above 100 is decisive evidence for the favourability of the Anisotropic model. Together with the Savage-Dickey result, this confirms that the data clearly favour a model that allows for separate treatment of radial and tangential uncertainties.

Conclusion

The anisotropic model consistently outperformed the isotropic model. Visually, it provided more accurate posterior predictive distributions for individual hole positions, capturing elliptical error structures consistent with physical intuition. Quantitatively, the Savage-Dickey ratio yielded a Bayes factor near zero in favour of the isotropic model, indicating negligible posterior mass at the isotropic subspace. Furthermore, nested sampling revealed a substantial log-evidence difference of $\Delta \log Z \approx 268$ in favour of the anisotropic model — decisively confirming its superiority.

Beyond statistical performance, the anisotropic model offers a more realistic representation of the measurement process: while radial positions are relatively well constrained by the curvature of the ring, tangential positions are more susceptible to misalignment and fragment distortion. This physical insight is reflected in the recovered posterior: $\sigma_r < \sigma_t$ with high confidence.

Taken together, our findings support a total hole count of approximately 355, consistent with a 354-day lunar calendar, and provide strong evidence that the measurement errors are direction-dependent. This highlights how carefully constructed statistical models — grounded in physical assumptions and evaluated through principled Bayesian methods — can yield valuable insights even from incomplete archaeological records.

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Appendix: Posterior Summary of Model Parameters

Anisotropic Model Posterior Summary

Parameter	Mean	Std	Median	5%	95%	n_eff	$\hat{R}$
N	355.10	1.27	355.10	352.98	357.17	150089.40	1.00
R	77.31	0.26	77.31	76.89	77.74	140073.39	1.00
phases[0]	-2.53	0.00	-2.53	-2.53	-2.52	266995.91	1.00
phases[1]	-2.14	0.00	-2.14	-2.14	-2.13	549871.36	1.00
phases[2]	-1.97	0.00	-1.97	-1.98	-1.97	169830.53	1.00
phases[3]	-1.32	0.01	-1.32	-1.35	-1.30	407464.84	1.00
phases[4]	-1.26	0.03	-1.26	-1.31	-1.21	376666.94	1.00
phases[5]	-1.25	0.01	-1.25	-1.26	-1.24	488787.59	1.00
σ_r	0.03	0.00	0.03	0.02	0.03	566747.07	1.00
σ_t	0.13	0.01	0.13	0.11	0.15	624075.38	1.00
x_offsets[0]	79.67	0.19	79.67	79.36	79.97	146941.56	1.00
x_offsets[1]	79.90	0.22	79.90	79.54	80.27	287612.22	1.00
x_offsets[2]	79.86	0.03	79.86	79.80	79.91	242810.23	1.00
x_offsets[3]	81.44	1.08	81.44	79.69	83.23	411059.14	1.00
x_offsets[4]	81.51	2.16	81.50	77.97	85.07	378157.08	1.00
x_offsets[5]	83.22	0.38	83.22	82.59	83.85	443628.60	1.00
y_offsets[0]	136.01	0.19	136.01	135.70	136.33	142977.35	1.00
y_offsets[1]	135.69	0.25	135.69	135.28	136.10	159439.48	1.00
y_offsets[2]	135.68	0.26	135.68	135.25	136.12	140196.16	1.00
y_offsets[3]	136.07	0.39	136.07	135.44	136.71	210488.10	1.00
y_offsets[4]	135.78	0.75	135.81	134.57	137.03	303370.16	1.00
y_offsets[5]	136.39	0.28	136.39	135.93	136.85	167609.76	1.00

Isotropic Model Posterior Summary

Parameter	Mean	Std	Median	5%	95%	n_eff	$\hat{R}$
N	354.50	2.79	354.50	349.91	359.11	189887.40	1.00
R	77.18	0.61	77.18	76.17	78.17	187735.89	1.00
phases[0]	-2.53	0.00	-2.53	-2.53	-2.52	437517.22	1.00
phases[1]	-2.14	0.01	-2.14	-2.16	-2.11	387101.74	1.00
phases[2]	-1.97	0.00	-1.97	-1.98	-1.97	236700.53	1.00
phases[3]	-1.32	0.05	-1.32	-1.40	-1.24	357285.89	1.00
phases[4]	-1.26	0.06	-1.26	-1.36	-1.16	339319.30	1.00
phases[5]	-1.25	0.02	-1.25	-1.29	-1.21	377427.05	1.00
σ	0.14	0.01	0.14	0.12	0.16	448359.76	1.00
x_offsets[0]	79.57	0.47	79.57	78.80	80.34	221776.87	1.00
x_offsets[1]	79.86	0.95	79.86	78.31	81.42	366747.92	1.00
x_offsets[2]	79.85	0.14	79.85	79.63	80.08	481017.58	1.00
x_offsets[3]	81.46	3.62	81.44	75.50	87.41	359572.52	1.00
x_offsets[4]	81.53	4.35	81.49	74.44	88.78	341364.77	1.00

x_offsets[5]	83.04	1.72	83.04	80.18	85.83	378210.55	1.00
y_offsets[0]	135.94	0.48	135.94	135.16	136.73	203864.59	1.00
y_offsets[1]	135.55	0.72	135.55	134.37	136.75	235688.08	1.00
y_offsets[2]	135.54	0.62	135.54	134.52	136.55	187794.31	1.00
y_offsets[3]	135.85	1.16	135.91	133.99	137.77	269183.78	1.00
y_offsets[4]	135.54	1.55	135.65	133.05	138.09	288180.62	1.00
y_offsets[5]	136.16	0.87	136.17	134.75	137.60	248778.41	1.00

Corner Plots from Nested Sampling

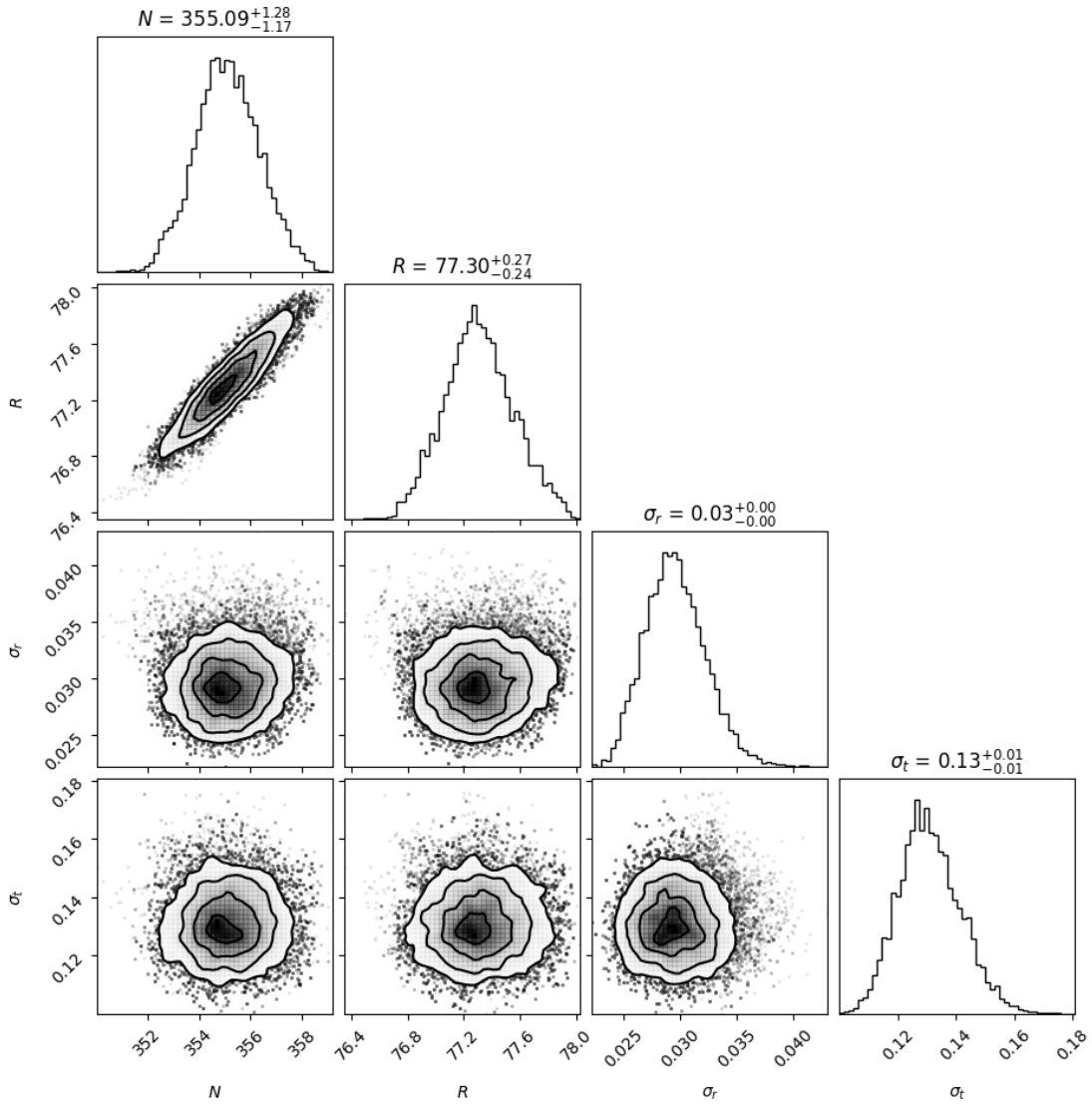


Figure 13: Corner plot from nested sampling for the anisotropic model, showing marginal and joint posterior distributions of key parameters.

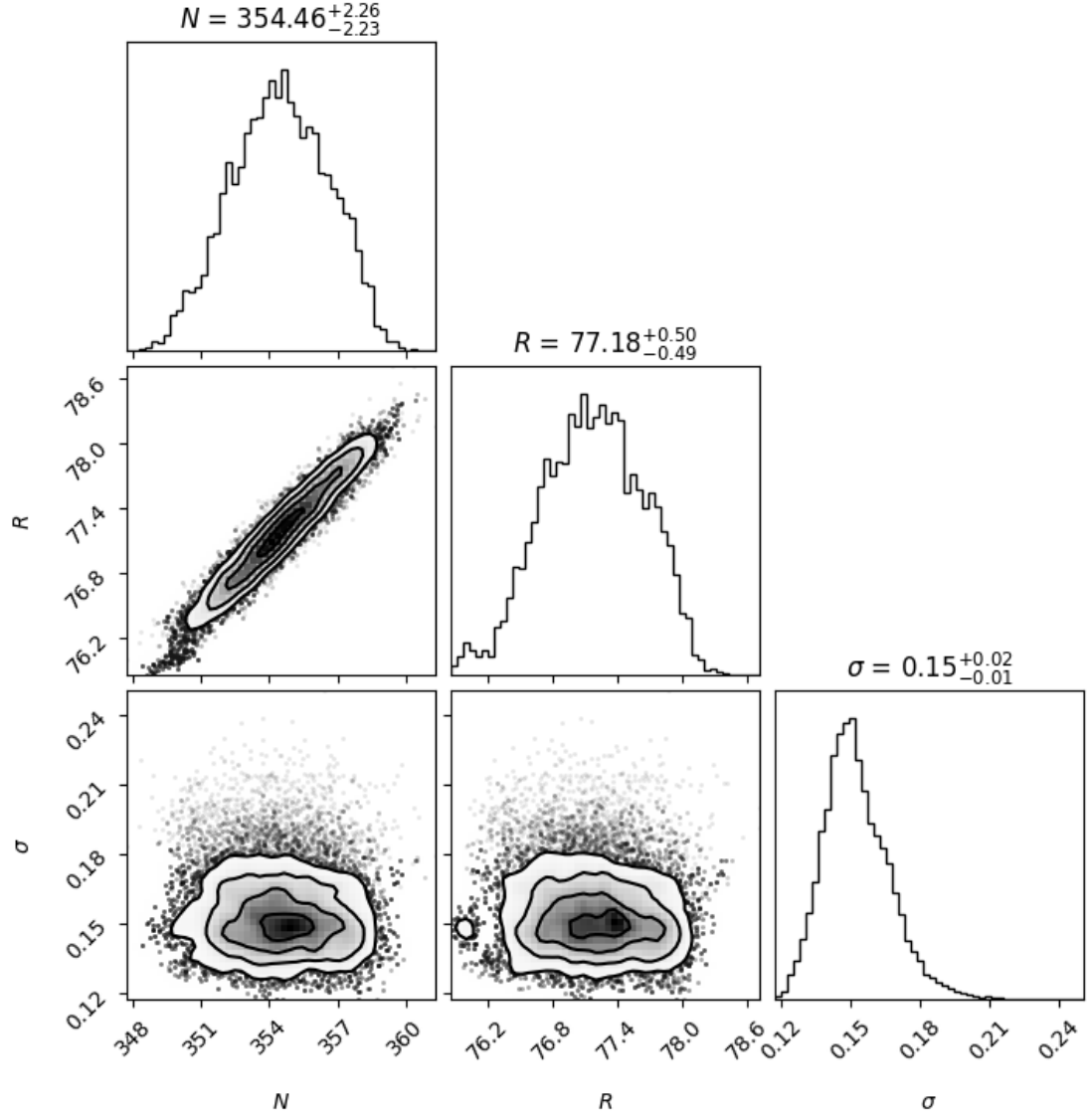


Figure 14: Corner plot from nested sampling for the isotropic model, showing marginal and joint posterior distributions of N , R , and σ .